Total employees
10

2. Sum() function
Write a SQL query to find the total salary paid to all employees in the IT department

```
Select sum('Salary') As 'total_Salary'
From employees
Where department = 'IT';
```

Department	Total_Salary
IT	220 000

3 AVG() Function
Write a SQL query to calculate the average salary of employees in the HR department.

```
Select AVG('Salary') AS 'Average Salary'
From employees
Where department = 'HR';
```

Department	Average_Salary
HR	49500

4 Min() and Max functions
Write a SQL query to find the highest and lowest salary in the company.

@ Select Min(Salary) As 'lowest Salary'
From employees;

Lowest_Salary
48 000

8 Select Max(Salary) As 'Highest Salary'
From employees;

Highest_Salary
62 000

5 GROUP BY Statement
Write a SQL query to group employees by department and display the total salary paid in each department.

Select Department, Sum('Salary') As 'total Salary'
From employees
GROUP BY 'Department';

Department	Total Salary
IT	220 000
finance	119 000
Marketing	105 000
HR	99 000

6 GROUP BY and COUNT()
Write a SQL query to count how many employees work in each city.

Select City, Count(*) As ~~employee~~ number of employees
From employees
GROUP BY City;

Number of employees
10

Number of employees	City
2	New York
3	Chicago
2	Los Angeles
2	San Francisco
1	Houston

7 GROUP BY and ORDER BY

Write a SQL query to group employees by department, calculate the average salary in each department, and order the results in descending order of average salary.

```
Select department, Avg(salary) As 'Average Salary'
from employees
GROUP BY 'department'
ORDERS BY 'Average Salary' Desc;
```

Department	Average_Salary
Finance	59500
IT	55 000
Marketing	52 500
HR	49 500

8 HAVING clause

Write a SQL query to find departments where the total salary paid exceedings 100000 (use GROUP BY and HAVING)

```
Select Department, Sum(Salary) As 'Total Salary'
from employees
GROUP BY 'Department'
Having Sum(Salary) > 100 000;
```

Department	Total_Salary
IT	220 000
finance	119 000
Marketing	105 000

9 Combining GROUP BY, HAVING and ORDER BY
Write a SQL query to list cities where more than one employee works, ordered by the number of employees in descending order.

```
Select City, COUNT(*) AS Number of employees
From employees
GROUP BY City
ORDER BY 'Number of employees' Desc
Having COUNT(*) > 1;
```

City	Number of employees
~~New York~~	
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

10 Combining Aggregate Functions
Write a SQL query to find the department with the highest average salary.

```
Select Department, Avg(`Salary`) As 'Average Salary'
From employees
GROUP BY 'department'
ORDER BY 'Average Salary' Desc
LIMIT 1;
```

Department	Average_Salary
Financ	59500.